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PALEOMAGNETIC DATA FROM TERTIARY VOLCANICS OF THE CAMPIDANO AND ASSOCIATED GRABENS, SARDINIA

M. MANZONI

Laboratorio di Geologia Marina del Consiglio Nazionale delle Ricerche, Bologna (Italy)

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67 samples of volcanic rocks from dykes and flows at the rims of the Campidano, Villamassargia and Fontanafredda grabens in southern Sardinia give a mean paleomagnetic direction of $D = 321.8$ and $I = +55.3$ with $\alpha_{95} = 8.9°$ after magnetic cleaning and correction for structural dip. The petrology and the age of the volcanics are not well documented; however, suggested ages vary between the upper Oligocene and the Pliocene. The new data are consistent with Sardinia having been part of the African plate since at least Permian times.

1. Introduction

The Campidano graben is the largest of the more recent structural features of Sardinia and crosses the island from its western coast to the southeast, thus separating two major blocks of Paleozoic, Mesozoic and crystalline rocks. The origin of the graben and its development have been studied recently by Cocozza and Schaefer [1]. In the area of Oristano the Campidano graben merges with another large foundered belt that runs from north to south along western Sardinia. A few minor transverse foundered valleys are also associated with the Campidano.

Widespread and composite volcanic activity has occurred at different times at the borders of the graben and within its floor [2–4]. Three major volcanoes are located at the northwestern end of the depression: M. Montiferru at the northern end, M. Arci in the eastern rim and M. Arcuentu at the western rim. Other volcanics, including trachytes, ignimbrites of different compositions, andesites and basalts, occurring in flows, domes, dykes and small plateaus, were formed between the Oligo-Miocene and the Pleistocene in different areas of the graben. A sketch map of the Tertiary volcanics associated with the Campidano and other grabens of Sardinia is shown in Fig. 1 [5].

2. Laboratory tests and results

Paleomagnetic data from fourteen sampling sites in the Tertiary volcanics are reported here; the exact geographical locations are given in the Appendix together with short petrographic descriptions. The suggested age of most of the volcanics sampled is Miocene and possibly Pliocene (basalts which are commonly assigned to the latest Plio-Pleistocene volcanic activity have not been sampled); however, no comprehensive petrological or even geological study of these volcanites has yet been published although the rock types have been referred to as basalts, andesites, andesitic porphyries, trachytes, rhyolites and acidic ignimbrites. The petrologic variation within the collection is remarkable but all seem to be related to the genesis of the graben.

From the fourteen numbered sampling sites (Fig. 1) 69 oriented hand samples were collected and cored, and 137 specimens measured. Two samples were excluded from the local mean calculations after measurement because they possessed strongly divergent directions of magnetization possibly because of sampling errors.

NRM directions (Table 1), in general, are not remarkably different from the directions measured after magnetic cleaning. NRM intensities vary from 4.2×10^{-4} gauss for the ignimbrites near Santumiali to 320

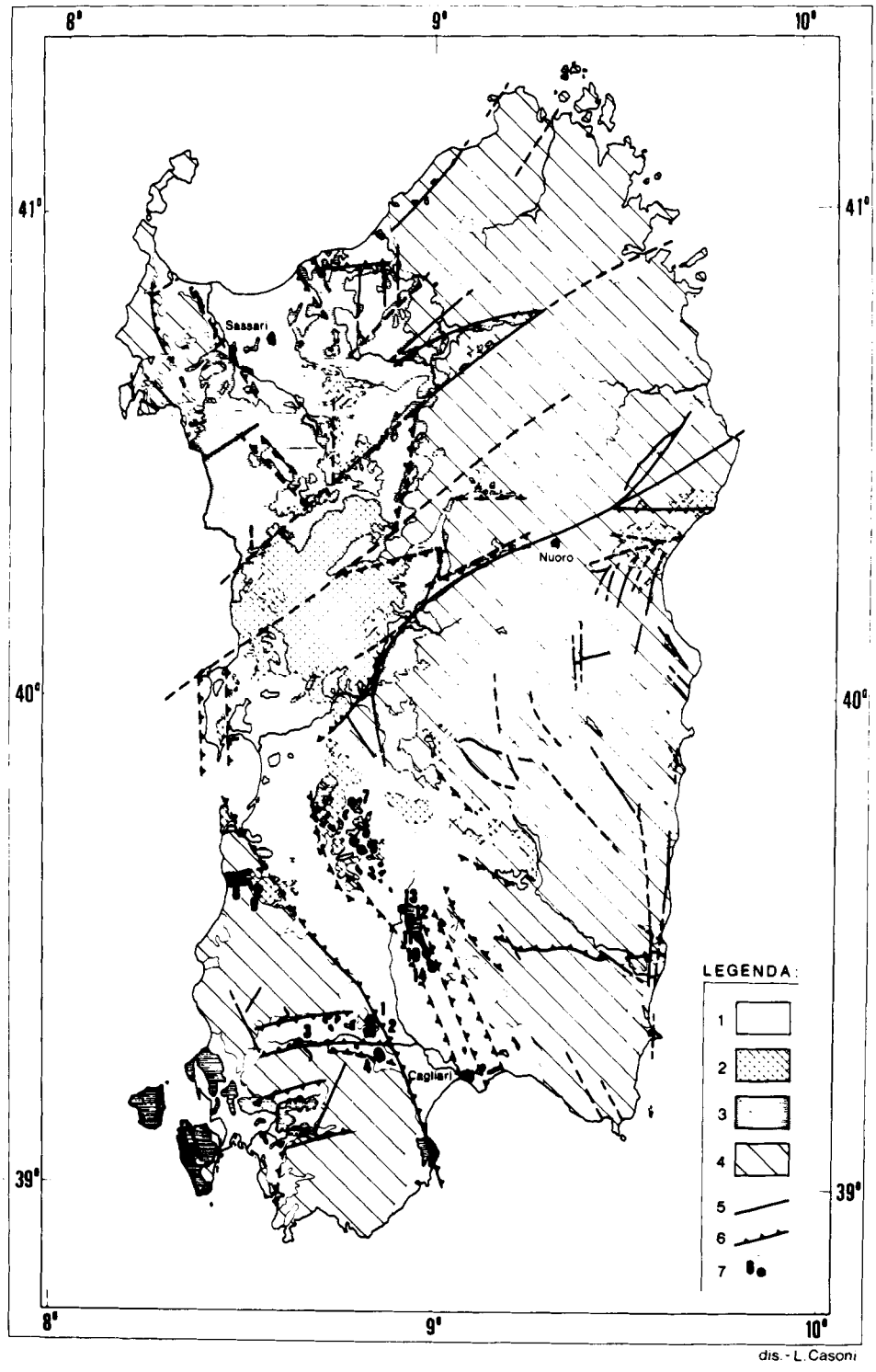


Fig. 1. Geologic sketch of Sardinia [5] with sampling locations. 1 = Tertiary and Quaternary basins, 2 = Plio-Pleistocene volcanics, 3 = Middle Tertiary volcanics, 4 = Pre-Tertiary basement and cover, 5 = faults, 6 = graben faults, 7 = sampling sites.

TABLE 1

Paleomagnetic results from 14 sites in volcanics from the Campidano area (southern Sardinia)

Locality number and name	N	n	NRM		M (10^{-4} gauss)	Oe	After cleaning		M (10^{-4} gauss)	R	K	α_{95}	Dip correction
			D	I			D	I					
1 Siliqua 1	5	10	44.1	+8.2	80.4	300	86	−10.3	20.2	4.90	39	22.3	
2 Siliqua 2	4	8	91	−8.1	50.6	300	85.9	−27.9	31.7	3.95	60	17.9	
3 Villamassargia	4	8	345.3	+64.1	146.0	150	329.1	+50.2	56.2	3.98	210	6.3	
4 Acquafredda	5	10	336.7	+59.3	197.0	200	334.6	+58.6	81.0	4.96	116	7.4	
5 Collinas	5	10	329.6	+55.5	63.3	200	330	+49.9	21.6	4.96	112	7.2	
6 Villanovaforru	5	10	317.9	+48.3	51.9	400	331.1	+58	3.53	4.72	14	20.9	*
7 Ales	5	10	6.2	+65	319.4	400	3.2	+66.5	3.59	4.99	325	4.2	*
8 Marina di Arbus	4	8	115.2	−30	36.1	400	118.3	−27.7	19.8	3.98	172	7.0	
9 Arcuentu	5	10	137.1	−46.7	71.1	400	119.8	−52.2	16.1	4.91	43	11.7	
10 Villagreca	5	10	329.4	+47.4	124.7	200	329.5	+43	47.6	4.96	109	7.3	*
11 Santumiali 1	5	10	343.6	+56.8	103.2	200	(0.5	+50.2)	79.7	4.96	107	7.4	*
12 Santumiali 2	5	10	308.8	+56	4.2	400	330.5	+63.6	3.0	4.95	81	8.5	*
13 Furtei	5	9	303.6	+65.3	201.5	200	307.2	+54.6	88.9	4.98	191	5.5	*
14 Monastir	5	10	103.5	−64.9	199.8	400	130.3	−69.5	97.2	4.97	125	6.8	
	67	133											
Sites mean (excl. sites 1,2,11	11						321.8	+55.3		10.63		8.9	
Sites mean (sites 6,10 12,13)	4						324.5	+55.2		3.94	12.5		

N = number of samples; n = number of specimens; M = magnetic moment; * = correction for dip applied.

$\times 10^{-4}$ gauss for the basalt dyke near Ales. One to three pilot specimens from every site were progressively demagnetized in AC fields up to 2,600 Oe maximum peak value. The decay of magnetic intensities with AC cleaning is shown in Fig. 2. The changes of the direction of magnetization with progressive AC cleaning of some pilot samples are shown in Fig. 3. The magnetization of the pilot samples from other sites is not plotted because directional changes with progressive demagnetization are less than 5° up to 1500 Oe maximum peak value. At site 11 (Santumiali I) a primary direction of magnetization cannot be established since the direction drifts continuously although slowly up to the highest applied field and a "hard" component of remanent magnetization is present. Another site with a relevant hard component is site 8 (Arbus). However, at this site the magnetic direction is stable.

Northwesterly declinations (southeasterly if reversed polarity) ranging from 285° to 335° east of north were found at ten sites. At two sites magnetic directions lie close to the present field, i.e. site 11, whose magnetization is not very stable, and site 8, where the highest correction for dip was applied because of the attitude of the beds induced by a nearby fault. Finally the remanent directions of the ignimbrites from Siliqua, sites 1 and 2, are different from any known paleomagnetic field direction from Sardinia.

Site mean inclinations vary between 10.3° and 69.5°, sites including both normal and reversed polarities. The mean inclination from all known Tertiary paleomagnetic data from Sardinia (110 sites) is 48°. Again, deviating inclinations are found at the sites 1, 2 (Siliqua ignimbrites with different declination), 7

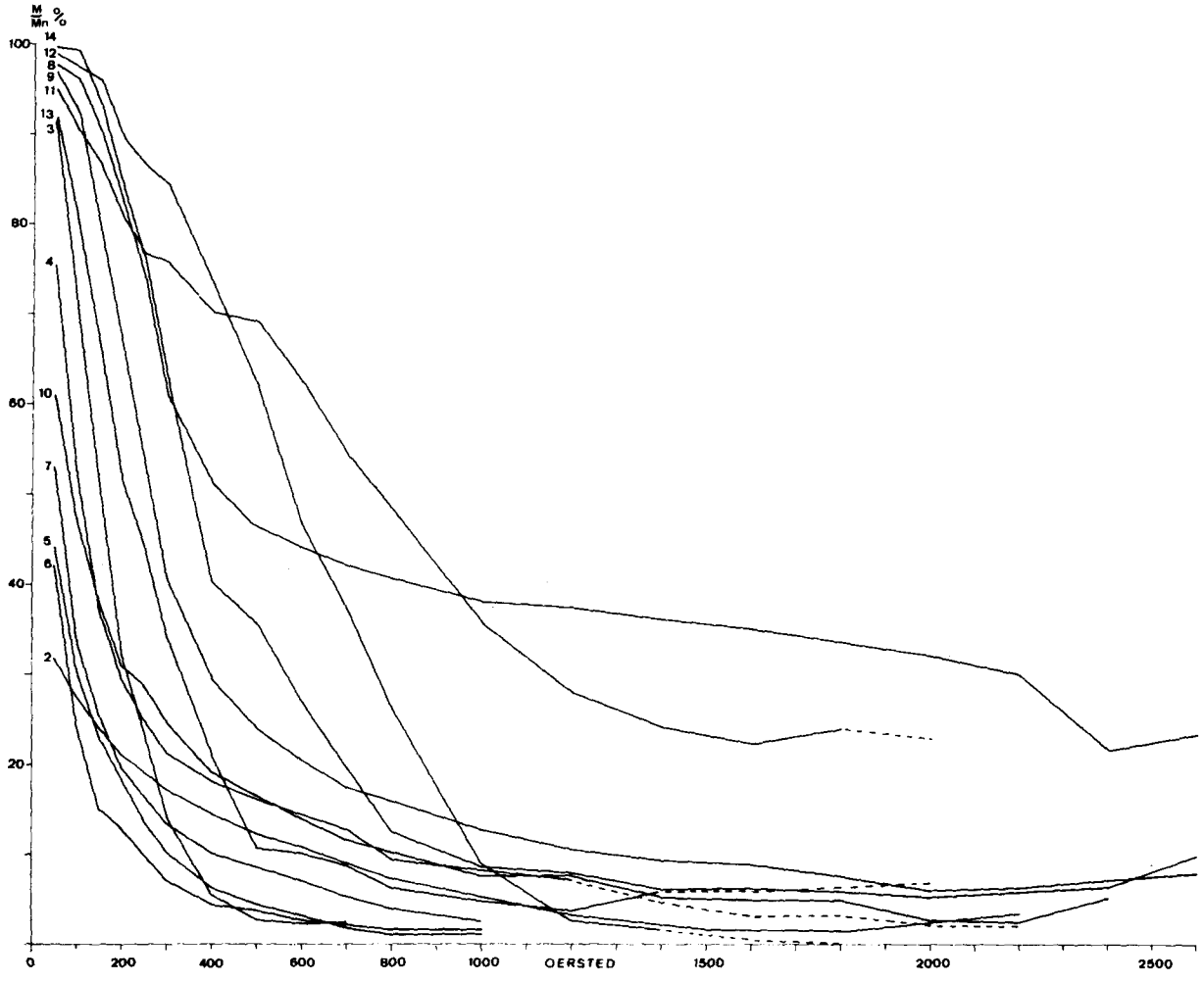


Fig. 2. Decay of magnetic intensities with progressive AC cleaning (percent of initial intensity of NRM).

(Ales with strong correction for dip), 8 (where a hard component of secondary magnetization is present), and 14. Fig. 4 shows directions of all samples. As in other parts of Sardinia, a large scatter between sites is found. Secular variation can account in part for this scatter but the possibility of major, unknown age differences (see discussion) leaves some doubt as to the validity of calculating an overall mean direction, and also of calculating pole positions for the Campidano region alone. Fig. 5 reports two mean directions after corrections for dip, where available: the first one averages all sites, excluding only the deviating data from Siliqua, and site 11 with unstable remanence; the second one is a "reference" mean direction including only those four sites (6,10,12,13) with precisely known tectonic control and stable magnetization. The two mean directions are only 3° apart from each other.

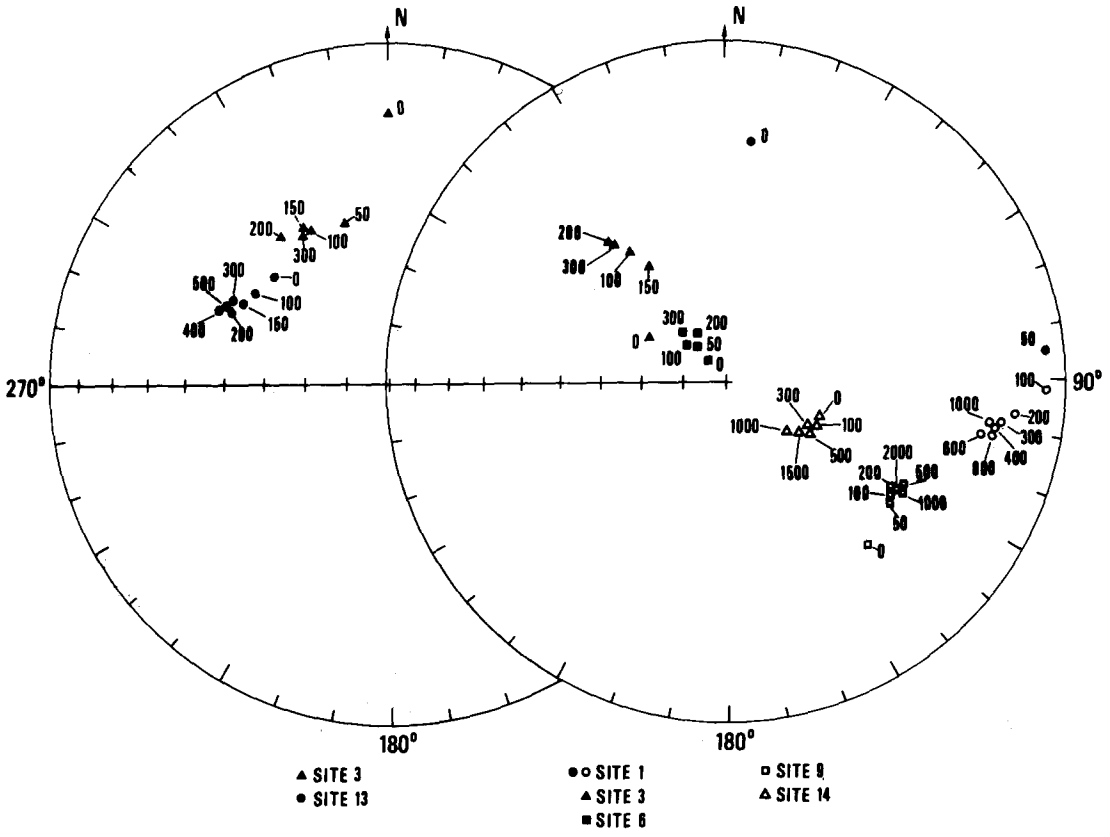


Fig. 3. Directional changes with progressive AC cleaning of pilot samples at sites 1, 3, 6, 9, 13, 14. Directional changes are negligible at the sites not shown in this diagram.

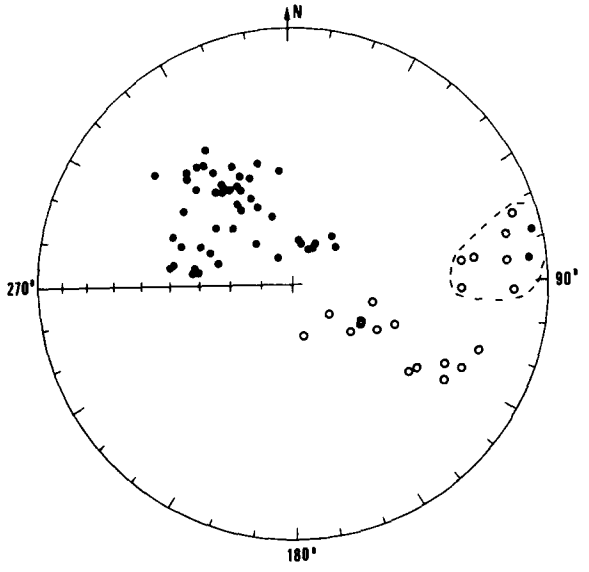


Fig. 4. Directions of magnetization of 67 samples from Campidano volcanics after AC cleaning and before corrections for dip. Directions from two outcrops near Siliqua lie within the dashed line.

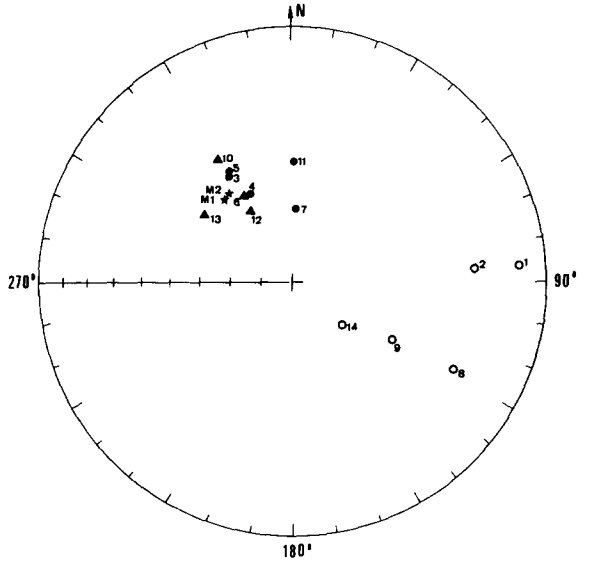


Fig. 5. Mean directions from 14 sites. Triangles = stable magnetization and correct tectonic control, M_1 = overall mean direction excluding sites 1, 2 and 11; M_2 = mean of sites 6, 10, 12, 13.

3. Discussion

The interpretation of the Campidano data is subject to the evaluation of a number of factors:

(1) *The ages* of the volcanics are not known with precision at the present time. Field relationships with the Miocene sediments of the eastern border of the graben suggest a lower age limit of Lower Miocene for most sites. Some of the units, for example the dykes of sites 5, 6 and 7, have been assigned in generic terms to "post-Helvetian" volcanic activity in the literature. The Arcuentu volcano (sites 8, 9) has been considered of Pliocene age mainly on geomorphological grounds, although recent data [6] suggest an older age. Age assignments for other sites are only tentatively referred to as "Upper Tertiary". Therefore the time-span of effusion of these volcanics can account for the occurrence of polarity reversals and the scatter of the paleomagnetic directions.

(2) *Secular variation* of the geomagnetic field may also be responsible for most of the scatter between the ten sites with northwesterly declinations since it is likely that a considerable time-span elapsed between the outpouring of chemically and petrologically different lavas in different parts of the graben complex.

(3) *The local tectonic setting* may well be responsible for the deviation of the directions of sites 1 and 2 (as in other cases in Sardinia); in fact the outcrop of conglomeratic ignimbrites of Siliqua, collected in an attempt to establish a conglomerate test, lies among alluvial deposits at the intersection of the western Campidano fault with the Villamassargia lateral graben. Most transverse faults of Sardinia have a component of strike-slip [5] which causes favorable conditions for rotations about a vertical axis of minor tectonic blocks especially where the movement is uninhibited at the intersection with the grabens. Therefore the Siliqua outcrops lie in an area of a complex, although concealed, tectonic situation. Strongly tilted beds also enclose the dyke of site 7. The volcanics of the remaining sites of Table 1, for which corrections for dip were not applied, are located in relatively undisturbed situations although the possibility of minor displacements cannot be ruled out or established with precision.

(4) The prevailing *northwestern declinations,* displaced 62° to 26° west of the present field, call for an explanation different from those based on secular variation or local tectonics. It is well known from previous paleomagnetic studies [7–10] that the northwestern direction of the field is characteristic of Permian and Tertiary Sardinian rocks, and that the pole positions calculated for Sardinia lie close to the African ones [11–13]. The change from northwestern to northern declinations was interpreted as the result of an anticlockwise rotation of Sardinia with respect to *stable* Europe (and not, as assumed by some writers, with respect to other Mediterranean lands), possibly from an earlier position closer to southern France [14,15].

4. Comparison with other data from Sardinia

Samples from a basalt flow of the Campidano graben floor near Uras, probably of Plio-Pleistocene age, were measured from the Campidano area [16]. These paleomagnetic directions cluster about the present field direction. Other data from Tertiary volcanics from southwestern Sardinia [17] are consistent with most Tertiary data from Sardinia (Fig. 6).

The majority of the mean site directions are oriented

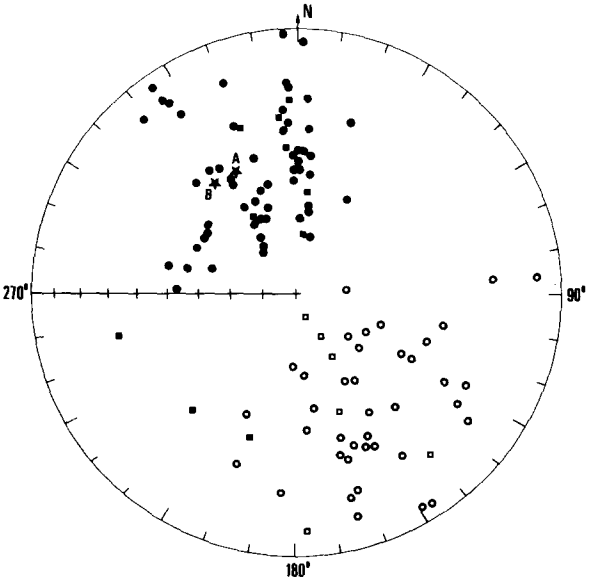


Fig. 6. Site means of published paleomagnetic data from the Tertiary volcanics of Sardinia, excluding the Plio-Pleistocene volcanics. Small squares: unreliable data or lacking precision parameters.

with northwesterly declinations (or southeasterly if reversed polarity). Directions aligned closer to the present field's normal direction are also frequently found, less frequently of reversed polarity [16–20]. The mean direction A of Fig. 6 includes all Tertiary sites, and the direction B excludes sites with paleodeclinations comprised between 35° and 10°.

At a few outcrops differently oriented declinations, to the northeast and to the west, were found [17,19,20]. Inclinations vary remarkably. It would appear that speculation on the form of the past geomagnetic field in Sardinia is not justified on the basis of these scanty and deviating data, and that their possible tectonic significance still needs an accurate assessment.

5. Conclusion

The new paleomagnetic data from Campidano are in good agreement with most Tertiary data for Sardinia and reinforce the general picture of "African" directions of the past geomagnetic field. Improved knowledge of the ages of the volcanics of the Campidano and its associated grabens would certainly help to establish the age of the transition of the paleomagnetic declinations from the northwest to the north and hence the age of significant geodynamic activity in the western Mediterranean Sea.

Appendix

Locations of the sampling sites

The rock names given here are found in the literature or in the local usage. The petrologic characteristics of the Campidano volcanics have not yet been described in a modern and comprehensive study. Longitudes are referred to M. Mario according to the I.G.M. topographic maps of Italy. Degrees and minutes are correct, seconds are estimated from the topographic maps.

Site 1. Conglomeratic ignimbrites on the road 1 km north of Siliqua. 39°18′40″N, 3°39′00″W.

Site 2. Conglomeratic ignimbrites 400 m north of Siliqua at a quarry on the road. 39°18′30″N, 3°38′50″W.

Site 3. "Andesites" with spheroidal jointing 2,5 km east of Villamassargia in a cut of the new road from Siliqua. 39°16′50″N, 3°47′20″W.

Site 4. "Andesitic porphyries" from a dome, at a cutting of the new road to Siliqua, 700 m north of the Acquafredda Castle. 39°17′20″N, 3°38′10″W.

Site 5. "Basalt" from a thick dyke in the main quarry at Collinas. 39°38′30″N, 3°37′10″W.

Site 6. "Andesite" from a dyke cutting Miocene sediments on the new road to Villanovaforru, about 2.5 km south of the town. 39°36′10″N, 3°34′30″W.

Site 7. "Basalt" from a dyke cutting Miocene sediments 400 m north of Ales, close to a major fault zone. It is debatable whether this dyke belongs to the Recent M. Arci volcano or to older Miocene activity. 39°46′10″N, 3°38′10″W.

Site 8. Acidic hyaloclastic tuffs on the road from Guspini 1 km northeast of the turning to Marina di Arbus. 39°37′00″N, 3°56′30″W.

Site 9. Acidic hyaloclastic tuffs on the main road to Marina di Arbus at Sa Mattoni. 39°35′50″N, 3°55′40″W.

Site 10. "Andesite" of the quarry northeast of Villagreca. 39°28′10″N, 3°16′10″W.

Site 11. "Green andesites at the dam near S. Miali. 39°32′20″N, 3°29′50″W.

Site 12. "Green trachytes" at the houses 500 m west of the dam at S. Miali. 39°32′10″N, 3°30′20″W.

Site 13. Ignimbrites on country road 400 m east of Furtei. 39°33′30″N, 3°30′10″W.

Site 14. "Basalts" from the largest quarry southeast of Monastir. 39°22′40″N, 3°23′50″W.

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